

KDLight: A Lightweight Knowledge Distillation Framework for Medical Image Classification

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Abstract—Conventional standalone approaches for diagnosing individual diseases often fail to achieve robust generalization because they are severely impacted by overfitting. This results in poor adaptability to diverse image representations and an inability to balance performance with computational efficiency. In this study, we propose KDLight, a lightweight, novel CNN model designed for efficient medical image classification across diverse modalities, including MRI, X-ray, radiography, skin images, and histopathology. We employ Knowledge Distillation (KD), where insights from an efficient teacher model (MobileNet) guide the learning process of the KDLight student model. The KDLight model minimizes the number of parameters while enhancing feature learning across diverse medical image representations. Experimental results show that KDLight achieves 95.55% classification accuracy with only 2.96 seconds and a compact 7.5 MB disk size, significantly reducing parameter size, accelerating inference, and lowering computational costs compared to traditional pre-trained models. Additionally, KDLight ability to efficiently learn diverse image representations can be extended to other domains, such as crack classification (e.g., road, window, and building cracks), enabling high-performance detection across different surface defect categories.

Index Terms—Knowledge Distillation, Lightweights, Multi-Modality, Generalize Learning

I. INTRODUCTION

Biomedical research and clinical diagnosis depend heavily on high-quality medical images, which are essential for accurate analysis and effective decision-making. The rapid increase in medical imaging data has made modalities such as X-ray, computed tomography (CT) scans images, magnetic resonance imaging (MRI), and Ultrasound crucial components of medical image retrieval systems [1]. Automated classification of these modalities is necessary for physicians to efficiently access relevant images by facilitating clinical education and

supporting diagnostic decision-making processes [2]. Medical image classification relies on Machine learning (ML) and Deep learning (DL) techniques to play a critical role in healthcare systems. It facilitates accurate diagnosis, effective treatment planning, and efficient disease monitoring [3].

Medical image classification is essential for diagnosing various types of diverse diseases across different imaging modalities. These imaging modalities provide critical insights into cellular structures and tumors, with tumor detection being challenging [4] due to complexity and cell overlapping from their small size. They enhance diagnostic accuracy and treatment outcomes by contributing significantly to advancements in healthcare systems. However, medical image classification still faces significant challenges that impact the accuracy and reliability of these systems by requiring ongoing advancements to overcome these limitations.

To obtain reliable findings, this [5] study presents a DL training framework that utilizes smaller models instead of larger ones, and introduces a model compression technique through knowledge distillation (KD) for leukocyte classification. Initially, 25,830 original images from the mixed leukocyte dataset are employed to pre-train large models based on CNN or Transformer architectures. Subsequently, the smaller models are trained using the distilled knowledge extracted from the pre-trained large models, capturing the essential features learned by the larger models.

While traditional DL models have demonstrated impressive performance in predicting single diseases, their application in multidisease prediction remains a significant challenge. CNN models are typically fine-tuned for specific diseases, often leading to overfitting on the unique features of a single condition. When tasked with predicting multiple diseases simultaneously, these models struggle to generalize effectively,

as the learned feature representations for one disease may not be transferable to others. This limitation hinders the model ability to accurately predict and differentiate between various medical conditions, underscoring the need for more optimal approaches that can handle the complexity and diversity inherent in multidisease prediction tasks. To overcome these limitations, the development of lightweight and efficient models has become critical. Methods based on knowledge distillation provide an effective solution by simplifying model architecture without compromising performance. Additionally, techniques such as model pruning [6] and quantization [7], [8] based offer alternative lightweight approaches that balance computational efficiency with accuracy. These approach not only reduces costs but also enhances patient with multidisease outcomes by enabling early intervention and personalized care.

Main contribution of this work as follows:

- **Unified Multi-Disease Diagnosis:** We propose KDLight, a unified framework capable of diagnosing multiple diseases simultaneously, overcoming the limitations of conventional models that focus on individual conditions.
- **Enhanced Generalization:** By leveraging knowledge distillation, KDLight transfers insights from a robust teacher model, improving its ability to classify a wide range of diseases effectively across complex medical scenarios.
- **Efficient and Scalable Design:** KDLight features a lightweight architecture that ensures high performance with faster inference times, making it suitable for real-time diagnostics in resource-limited healthcare settings and applicable to diverse domains, such as surface defect detection.

II. PRIOR WORK

This section presents research that combines techniques for diagnosing multiple medical conditions through image classification, with an emphasis on balancing high performance and computational efficiency.

A. Techniques for Multi-Disease Medical Image Classification

Chen et al. [9] proposed a lightweight CNN using knowledge distillation for cervical cancer classification. The Xception model achieved 72.25% accuracy on the Herlev Pap smear 5-class dataset, compared to the Inception-ResNetV2 model, which achieved 73.58% accuracy. However, the model's generalizability to diverse imaging modalities remains a limitation by restricting its applicability to varied medical datasets. El-Assiouti et al. [10] proposed a Hybrid Data-efficient Knowledge Distillation (HDKD) network for medical image classification. It employs a CNN teacher and a hybrid student, using the Mobile Channel-Spatial Attention (MBCSA) block as the core component. The method achieved 92.15% accuracy on the skin disease HAM-10000 dataset for 7-class classification. However, its limited generalizability to diverse imaging modalities restricts its applicability to other medical datasets. Leng et al. [5] proposed a DL framework using knowledge distillation (KD) for leukocyte classification. Large CNN and Transformer models were pre-trained on a 5 classes blood cells images

dataset and their distilled knowledge trained smaller models. Their best model achieved 98.31 accuracy but showed limited generalizability to diverse imaging modalities by restricting its broader applicability. Sepahvand et al. [11] proposed a lightweight knowledge distillation model for classifying breast cancer histopathological images from the BreakHis binary class dataset. It trains two student models, based on VGG and ResNext teacher models, using adaptive joint learning to transfer dark knowledge from both the final-layer outputs and middle-layer feature maps. The ResNeXt-based student model achieved 97.09% accuracy with reduced parameters, though its limited generalizability to multi-class classification impacts its effectiveness in diverse scenarios. Termritthikun et al. [12] proposed a knowledge distillation (KD) strategy for efficient multi-label Chest X-ray (CXR) classification. By using CNNs and Transforms as teacher models, they distilled knowledge to a smaller student model by achieving high performance with fewer parameters. Their approach was tested on ChestX-ray14 dataset and achieved AUC scores of 83.70% by making it suitable for resource-constrained devices.

B. Balancing Accuracy and Computational Efficiency in Medical Diagnostics

Zhou et al. [13] proposed the Spiking Cortical Model-based Global and Local (SCM-GL) attention module to enhance light-weight CNNs by combining local and global attention masks to highlight key features. It was evaluated on brain MRIs, chest X-rays, and osteosarcoma images by demonstrating superior performance over other attention mechanisms. Jubair et al. [14] proposed a lightweight CNN leveraging EfficientNet-B0 as a TL model for binary classification of oral cancer and achieved 85% accuracy. However, its binary focus limits generalizability to multiclass oral lesion cases. Wang et al. [15] proposed SSD-KD, a knowledge distillation method designed to improve lightweight models for skin disease classification under resource constraints. The method was evaluated on the ISIC 2019 dataset and achieved 85% accuracy in classifying 8 skin diseases using dermoscopic images. Asham et al. [16] proposed a lightweight DL model for detecting pulmonary diseases 4-class dataset. Their method integrates knowledge distillation and the ConvMixer block within a MobileNet-based student model. It achieved 97.92% accuracy on chest X-ray datasets with 0.63 million parameters by making it suitable for resource-limited settings. Ahmadi et al. [17] proposed a lightweight knowledge distillation method for COVID-19 detection. The approach utilized VGG19 and ResNet50V2 as teacher networks and MobileNetV2 as the student network. Their model was tested on normal and COVID-19 cases binary class dataset and achieved an accuracy of 98.8%. Although effective for mobile and embedded devices but may perform poorly on multi-class and diverse modality datasets.

III. PROPOSED APPROACH

This section will comprehensively cover the study materials and methodology. Section A will explain the novel KDLight

model with the distillation of teacher knowledge. Section B will provide an overview of the proposed KDLight student model. In Section C, we will outline the KDLight learning process, focusing on enhancing various medical image features. Finally, the hyperparameter settings for both the teacher and student models will be presented at the conclusion of the study.

A. KDLight: A Lightweight Distilled Teacher Knowledge Framework

The scale and complexity of neural models present a significant barrier, particularly when it comes to deploying them on edge devices with limited resources, which we address in this section. We chose MobileNet, a well-established model with strong capabilities and a lightweight architecture, as our teacher model for this study. Simultaneously, we developed KDLight, a lighter network, to act as our student model. Despite the advantages of a simplified architecture, the student model may not perform as well as its larger counterparts. As described in [18], we employed Knowledge Distillation (KD) techniques to bridge this performance gap. By applying this method, we were able to maintain the student model's lower processing requirements while enhancing its performance. Several pre-trained student architectures were used in our study, some of which had as many parameters as, or even more than, our student model. This investigation allowed us to evaluate various performance indicators for each architecture used as a student model, providing valuable insights into the effectiveness of KD in our case. The complete KD approach used in this study is illustrated in Fig. 1.

In the context where MobileNet is the teacher model and KDLight is the student model, the Knowledge Distillation (KD) equations in this research expressed as:

$$\mathcal{L}_{KD} = \alpha \mathcal{L}_{CE}(y, \hat{y}) + \beta \mathcal{L}_{KD}(q_{MobileNet}, \hat{q}_{KDLight}) \quad (1)$$

$\mathcal{L}_{CE}(y, \hat{y})$ represents the cross-entropy loss between the true labels y and the predicted labels $\hat{y}$ from the KDLight student model. $\mathcal{L}_{KD}(q_{MobileNet}, \hat{q}_{KDLight})$ denotes the distillation loss between the softened output probabilities $q_{MobileNet}$ from the MobileNet teacher model and the softened output probabilities $\hat{q}_{KDLight}$ from the KDLight student model. Here, α and β are hyperparameters that control the relative importance of the cross-entropy loss and the distillation loss. The subscripts CE and KD refer to cross-entropy and knowledge distillation, respectively. Additionally, q represents the softmax output distributions.

Softening the outputs: The softened probability distributions for both models are generated using temperature scaling. The temperature hyperparameter T in equations (2) and (3) is used to control the softness of the probability distributions. Specifically, it is applied to the logits $t_{MobileNet,i}$ and $s_{KDLight,i}$ from the MobileNet teacher and KDLight student models, respectively. By setting T to a value greater than 1, the probability distributions are softened, making the model less confident in its predictions. This helps in transferring knowledge from the teacher to the student model. The $t_{MobileNet,i}$

and $s_{KDLight,i}$ are the pre-softmax outputs from the teacher and the student models, respectively.

$$q_{MobileNet,i} = \frac{\exp(t_{MobileNet,i}/T)}{\sum_j \exp(t_{MobileNet,j}/T)} \quad (2)$$

$$\hat{q}_{KDLight,i} = \frac{\exp(s_{KDLight,i}/T)}{\sum_j \exp(s_{KDLight,j}/T)} \quad (3)$$

In this formulation: The MobileNet model, acting as the teacher, provides softened output logits $t_{MobileNet}$, guiding the KDLight student model to learn from this richer representation. The KDLight student model, with logits $s_{KDLight}$, learns both from the true labels via cross-entropy and from the softened outputs of the teacher via distillation loss, improving its performance on multidisease classification tasks while maintaining its lightweight architecture.

B. Proposed Novel KDLight Student Model

In this section, we present the architectural details of the proposed novel model. The KDLight Student model follows a typical CNN architecture, designed to progressively extract and refine features for multidisease medical image classification. The model starts with an input layer accepting images of size 224 x 224 x 3, followed by several convolutional layers (Conv2D) with ReLU activations to capture hierarchical feature patterns. Each convolutional block is accompanied by max-pooling layers that progressively downsample the spatial dimensions, reducing the complexity while retaining critical features. Dropout layers (ranging from 0.25 to 0.5) are strategically placed throughout the network to prevent overfitting by randomly dropping units during training.

The network deepens through successive convolutional layers, increasing the depth of feature maps from 32 to 128 channels. Notably, after the 6th convolutional layer, the depth of feature maps reduces from 128 to 16. After the convolutional and pooling layers, the 10 x 10 x 128 feature map is flattened into a 6400-dimensional vector. This is followed by fully connected dense layers (128 and 64 units) with ReLU activations, batch normalization, and dropout layers to enhance generalization and robustness. The architecture enables the KDLight model to efficiently learn complex features across diverse medical conditions, ensuring high performance while maintaining efficiency. The final design also leverages regularization techniques to avoid overfitting, ensuring the model robustness for real-world applications. Table I illustrated the overall layer by layer architecture of novel KDLight model. To optimize training, we selected the Adam optimizer, with an initial learning rate of 1e-3. The Adam optimizer was chosen for its adaptive learning rate mechanism, which facilitates faster convergence compared to other optimizers. This learning rate is commonly adopted in medical image classification, as evidenced by recent studies published in high-impact journals. A batch size of 32 was used by striking an effective trade-off between computational efficiency and model performance. We trained the models for 20 epochs, a number is determined by monitoring the validation performance, where we observed

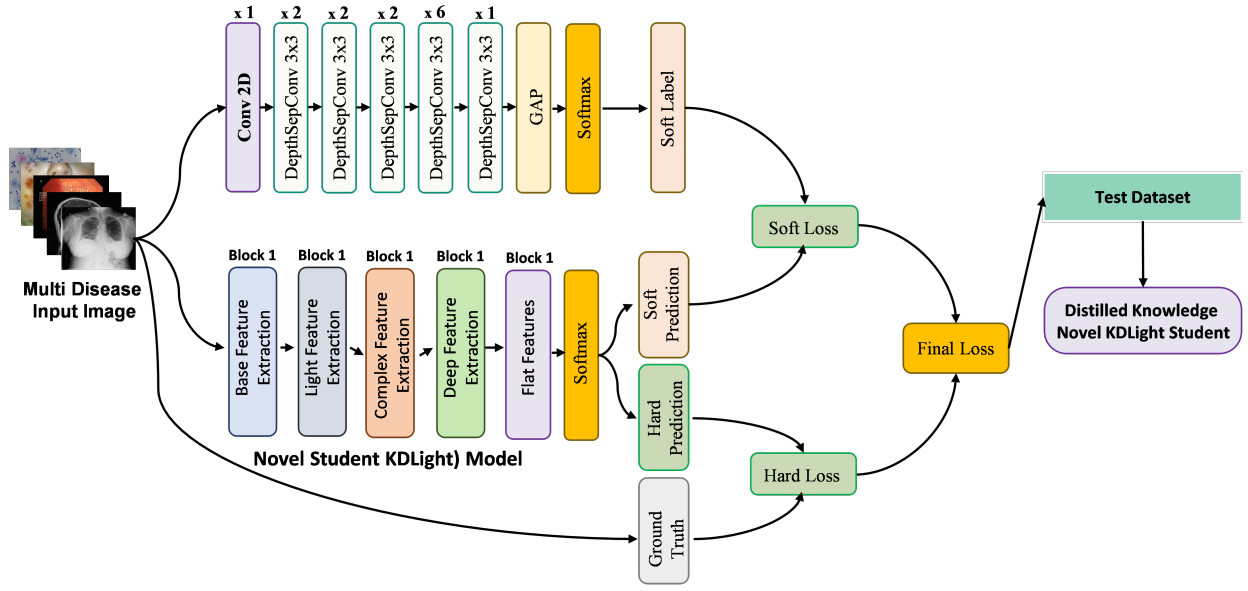


Fig. 1. Overview of Novel KDLight with Distillated Teacher Knowledge Framework

TABLE I
PROPOSED KDLIGHT MODEL NETWORK LAYER BY LAYER OVERVIEW

Layer name	Layer type	Output shape
Input	Input layer	224 x 224 x 3
Conv2d 1st	Conv2D layer	222 x 222 x 32
Activation Function	ReLU layer	222 x 222 x 32
Max Pooling	Max Pooling 2d	111 x 111 x 32
Conv 2nd	Conv2D layer	109 x 109 x 32
Activation Function	ReLU layer	109 x 109 x 32
Max Pooling	Max Pooling 2d	54 x 54 x 32
Dropout (0.25)	Dropout	54 x 54 x 32
Conv2d 3rd	Conv2D layer	52 x 52 x 64
Activation Function	ReLU layer	52 x 52 x 64
Conv2d 4th	Conv2D layer	50 x 50 x 64
Activation Function	ReLU layer	50 x 50 x 64
Max Pooling	Max Pooling 2d	25 x 25 x 64
Dropout (0.25)	Dropout	25 x 25 x 64
Conv2d 5th	Conv2D layer	23 x 23 x 128
Activation Function	ReLU layer	23 x 23 x 128
Conv2d 6th	Conv2D layer	21 x 21 x 128
Activation Function	ReLU layer	21 x 21 x 128
Max Pooling	Max Pooling 2d	10 x 10 x 128
Dropout (0.25)	Dropout	10 x 10 x 128
Conv2d 7th	Conv2D layer	10 x 10 x 16
Conv2d 8th	Conv2D layer	10 x 10 x 16
Activation Function	ReLU layer	10 x 10 x 16
Activation Function	ReLU layer	10 x 10 x 16
Max Pooling	Max Pooling 2d	10 x 10 x 128
Flatten	Flatten layer	6400
Dense	Dense Layer	128
Activation Function	ReLU layer	128
Batch Norm	Batch Norm	128
Dropout (0.25)	Dropout	128
Dense	Dense Layer	64
Activation Function	ReLU layer	64
Batch Norm	Batch Norm	64
Dropout (0.5)	Dropout	64

minimal improvements beyond this point. To prevent overfitting, training was terminated after 20 epochs. During training, we employed Model Checkpoint to save the best-performing

models and used Early Stopping to terminate training when performance decayed.

IV. EXPERIMENTS

This section provides a detailed overview of the experimental setup, including dataset collection, and performance metrics such as accuracy, precision, recall, and F1 score. We compare the teacher model, MobileNet, with the proposed KDLight student model, evaluating their performance across different classes and assessing computational efficiency. An ablation study further highlight the strengths and limitations of the models.

A. Experimental Settings

We conducted all model training using TensorFlow and Keras within our experimental framework, leveraging the computational resources provided by Google Colab. The training process was carried out on an NVIDIA Tesla K80 GPU with 12GB of RAM. In this study, we employed various statistical evaluation metrics including accuracy, precision, recall, and F1-score, to evaluate the performance of the proposed model. The calculations for these metrics are derived from the confusion matrix, which encompasses the components of true positives, true negatives, false positives, and false negatives. The specific formulations used to derive these metrics are detailed in equations (4) through (7).

$$\text{Precision} = \frac{\text{True Correct}}{\text{True Correct} + \text{True Incorrect}} \quad (4)$$

$$\text{Recall} = \frac{\text{True Correct}}{\text{True Correct} + \text{False Incorrect}} \quad (5)$$

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{Accuracy} = \frac{\text{True Correct} + \text{False Correct}}{\text{True Correct} + \text{True Incorrect} + \text{False Correct} + \text{False Incorrect}} \quad (7)$$

B. Dataset Collection

This section provides detailed information about the datasets used to evaluate the efficacy and reliability of our proposed model in distinguishing between various medical conditions across different modalities. The datasets include MRI [19], chest X-ray [20], endoscopic [4], and cell [4] images. Each modality offers distinct features, such as infection regions, which are crucial for evaluating the model's performance in diverse medical contexts. For dataset distribution, we have used [21], [22] 70% of the data for training the model, 10% for validation, and 20% for testing. Table II summarizes the distribution of datasets across multiple classes for training, validation, and testing.

TABLE II
DATASET DISTRIBUTION ACROSS TRAIN, VALIDATE, AND TEST SET

Disease	Train	Validate	Test	Total
BA-impetigo	72	8	20	100
Colorectal cancer	100	11	28	139
COVID	1440	160	400	2000
Esophagitis	77	9	21	107
Glioma	1188	132	300	1620
High Squamous	117	13	33	163
Low Squamous	81	9	23	113
Lung Opacity	1440	160	400	2000
Meningioma	1206	133	306	1645
Negative	441	48	123	612
Pituitary	1311	146	300	1757
Pylorus	108	12	30	150
SCC	53	6	15	74
VI-Chickenpox	122	14	34	170
Viral Pneumonia	968	108	269	1345

C. Performance Evaluation of Teacher (MobileNet) and Proposed Novel KDLight

In this section, we evaluate the performance of the teacher model (MobileNet) and the proposed KDLight student model, as shown in Table III. The teacher model achieved an overall accuracy of 96.56%, with excellent precision scores of 1.00 for classes like VI-chickenpox and Viral Pneumonia. The recall for BA-impetigo and Colorectal Cancer was perfect (1.00), while the recall for Esophagitis (0.86) and Viral Pneumonia (0.87) were the lowest. F1-scores for Viral Pneumonia (0.99) were among the highest, with BA-impetigo and Negative close behind. In contrast, the student model (KDLight) achieved an accuracy of 95.22%. It showed perfect precision for BA-impetigo, High Squamous, COVID, Pituitary, and Pylorus, but the Glioma class had the lowest precision at 0.74. Recall scores for BA-impetigo, COVID, and Lung Opacity were perfect, while Negative had the lowest recall (0.73). F1-scores for BA-impetigo, High Squamous, and COVID were perfect, while Negative and Glioma exhibited the lowest scores, signaling areas for improvement.

Some classes, like Esophagitis and Glioma, may be inherently more complex or similar to other diseases in terms

of their visual features. This overlap can make it harder for the model to distinguish between them accurately, resulting in lower precision and recall values. The confusion matrix is an essential for evaluating classification models, providing a detailed comparison of true labels and predicted labels. Fig 2 illustrates the confusion matrix for the teacher model (MobileNetV1), while Fig 3 presents the student model (Novel KDLight). In Fig 2, the teacher model classified 2,223 out of 2,302 test images correctly across 15 disease classes, with 79 misclassifications. In contrast, Fig 3 shows that the student model correctly classified 2,192 samples, resulting in 110 misclassifications. Despite the increase in errors, the Novel KDLight model demonstrated lower computational costs, showcasing its potential for deployment in resource-constrained environments. The ROC-AUC curve visually represents a model's performance by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different threshold settings. The Area Under the Curve (AUC) measures the model's ability to distinguish between positive and negative classes, with higher values indicating better performance. As shown in Fig 4, the Teacher model (MobileNetV1) achieved an impressive AUC of 0.9990, while the Novel KDLight Student model had a slightly lower AUC of 0.9985. Despite the marginal difference, the Student model offers significant advantages with its lightweight architecture, making it more computationally efficient for real-world deployment. Fig 5 presents the precision-recall (PR) curve, where the Teacher model recorded an Average Precision (AP) of 0.9865, compared to 0.9819 for the Student model. This small difference suggests that both models perform similarly, but the Teacher model slightly outperforms the Student model in precision and recall.

D. Performance Evaluation of KDLight with existing models

The performance comparison of Novel KDLight with existing pre-trained models (as shown in Table IV) demonstrates that Teacher MobileNet model consistently outperform DenseNet, ResNet, VGG, Inception, and Xception series across multiple metrics. MobileNetV1 achieves the highest accuracy at 96.55%, outperforming the InceptionV3-Base model (90.80% accuracy). Notably, the proposed KDLight model outperform even Teacher MobileNetV1, achieving 95.22% accuracy. This underscores KDLight ability to deliver superior performance for multidisease medical classification tasks, combining high precision, efficient learning, and robustness in real-world applications.

E. Complexity analysis of Teacher (MobileNet) and Proposed Novel KDLight student

The complexity analysis, shown in Table V, compares the teacher model and the novel KDLight student model in terms of computational efficiency using floating-point operations per second (FLOPs). The teacher model demands 73.4 billion FLOPs, while the KDLight model significantly reduces this to 5.15 billion FLOPs, showcasing its lightweight design. Additionally, the KDLight model uses only 7.5 MB of memory

TABLE III
CLASS-WISE PERFORMANCE EVALUATION OF TEACHER AND PROPOSED NOVEL KDLIGHT STUDENT MODEL

Model	Classes	Performance Metrics			
		Precision	Recall	F1-Score	Accuracy
Teacher (MobileNet)	BA-impetigo	0.95	1.00	0.98	96.56%
	Colorectal cancer	0.93	1.00	0.97	
	COVID	0.97	0.96	0.96	
	Esophagitis	0.95	0.86	0.90	
	Glioma	0.98	0.97	0.97	
	High Squamous	0.94	0.91	0.92	
	Low Squamous	0.81	0.96	0.88	
	Lung Opacity	0.96	0.97	0.97	
	Meningioma	0.95	0.97	0.96	
	Negative	0.97	0.98	0.98	
	Pituitary	0.99	0.98	0.98	
	Pylorus	0.93	0.93	0.93	
	SCC	0.95	0.96	0.97	
	VI-chickenpox	1.00	0.87	0.80	
	Viral Pneumonia	1.00	0.97	0.99	
Student (Novel KDLight)	BA-impetigo	1.00	1.00	1.00	95.22%
	Colorectal cancer	0.95	0.90	0.92	
	COVID	1.00	1.00	1.00	
	Esophagitis	0.81	0.81	0.81	
	Glioma	0.74	0.79	0.76	
	High Squamous	1.00	1.00	1.00	
	Low Squamous	0.90	0.96	0.93	
	Lung Opacity	0.95	1.00	0.98	
	Meningioma	0.87	0.87	0.87	
	Negative	0.78	0.73	0.71	
	Pituitary	1.00	0.97	0.99	
	Pylorus	1.00	0.98	0.99	
	SCC	0.97	0.97	0.97	
	VI-chickenpox	0.96	0.96	0.96	
	Viral Pneumonia	0.98	0.98	0.98	

TABLE IV
PERFORMANCE EVALUATION OF KDLIGHT AND TWELVE PRE-TRAINED MODEL

Reference	Method	Accuracy
Gao Huang et al [23]	DenseNet121	93.90
Gao Huang et al [23]	DenseNet169	94.88
Gao Huang et al [23]	DenseNet201	94.54
Szegedy et al [?]	InceptionV3	89.32
Howard et al [24]	MobileNetV1	96.55
Howard et al [24]	MobileNetV2	94.68
He et al [26]	ResNet50V2	92.22
He et al [26]	ResNet101V2	92.27
He et al [26]	ResNet152V2	92.86
Simonyan et al [25]	VGG16	94.34
Simonyan et al [25]	VGG19	92.12
Simonyan et al [25]	Xception	90.80
Proposed	KDLight	95.22

compared to the teacher's 21.2 MB, and its inference time is reduced from 17.56 seconds to just 2.96 seconds. These results highlight the KDLight model's superiority in memory efficiency and speed, making it ideal for resource-constrained, real-time applications. The analysis emphasizes the trade-off between performance and efficiency, demonstrating how the KDLight model can achieve near teacher model performance while greatly reducing computational overhead.

F. Ablation study

Ablation studies are crucial to evaluate the impact of individual components within a model by ensuring performance

TABLE V
COMPLEXITY ANALYSIS OF TEACHER AND PROPOSED NOVEL KDLIGHT STUDENT MODEL

Model	Memory	Flops	Test Time
Teacher (MobileNet)	21.2 MB	73.4 Billion	17.56 seconds
Student (Novel KDLight)	7.5 MB	5.15 Billion	02.96 seconds

gains are directly attributed to the proposed changes. Table VI demonstrates significant improvements after integrating the KDLight Student Model, with precision increasing from 88.75% to 96.60%, recall from 84.73% to 93.23%, and F1-Score from 85.75 to 90.49, indicating better balance and robustness. Accuracy also improved from 94.61% to 95.22%, confirming the effectiveness of the KDLight model in enhancing overall performance with Knowledge distillation technique. The down arrow shows decline trend in performance and up arrow indicate rise in performance. We compare the

TABLE VI
PERFORMANCE EVALUATION BEFORE AND AFTER NOVEL KDLIGHT STUDENT MODEL

Model	Precision	Recall	F1-Score	Accuracy
Before KD	88.75	84.73	85.75	94.61 ↓
After KD	96.60	93.23	90.49	95.22 ↑

performance of the proposed KDLight model with different teacher models instead of MobileNet to evaluate its effectiveness in terms of recall and accuracy. The results of this

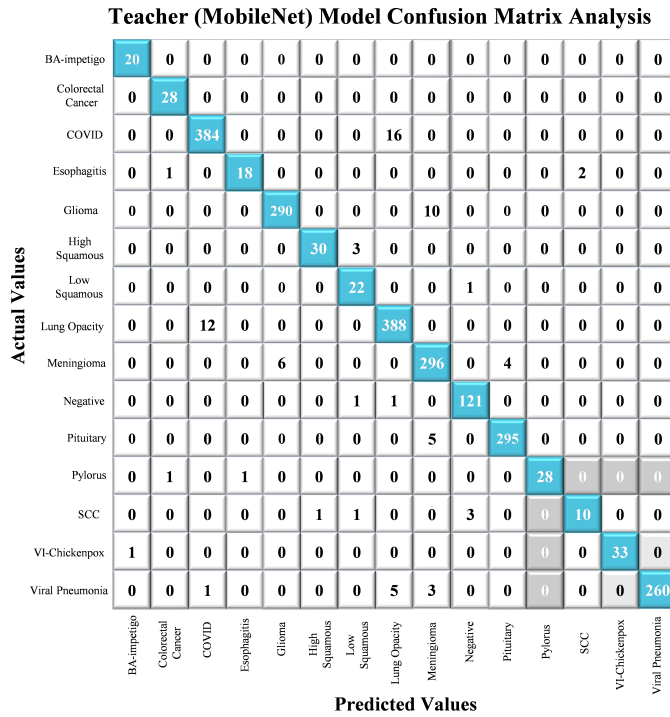


Fig. 2. Confusion matrix visualization of Teacher Model

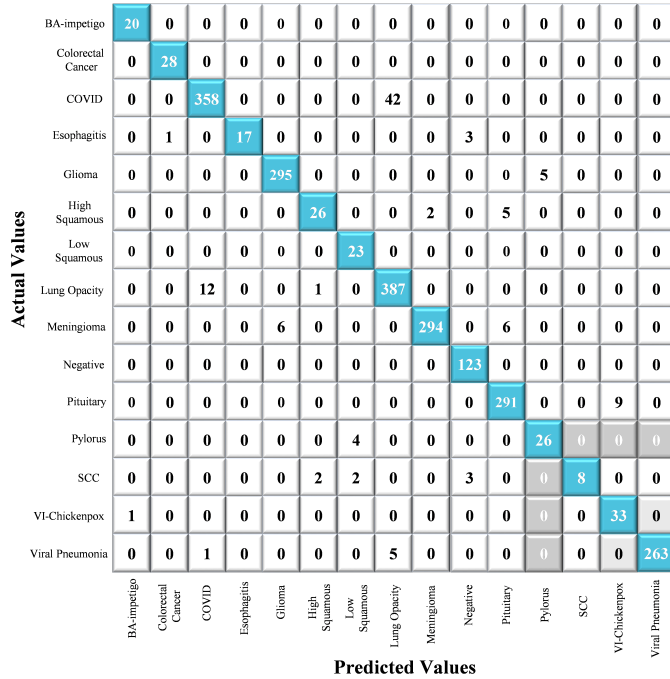


Fig. 3. Confusion matrix visualization of Student (Novel KDLight) Model

comparison are presented in Table VII, where the lowest-performing model is KDLight with DenseNet121-Base. This model achieves the recall rate of 89.12% and an accuracy of 92.27%, which are the lowest among all tested teacher models. In contrast, the highest-performing model is KDLight

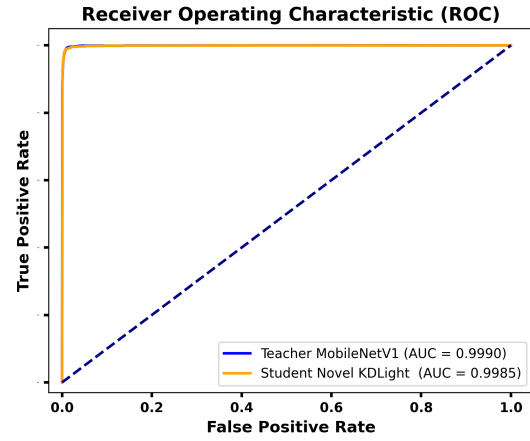


Fig. 4. ROC-AUC curve visualization for multi medical disease

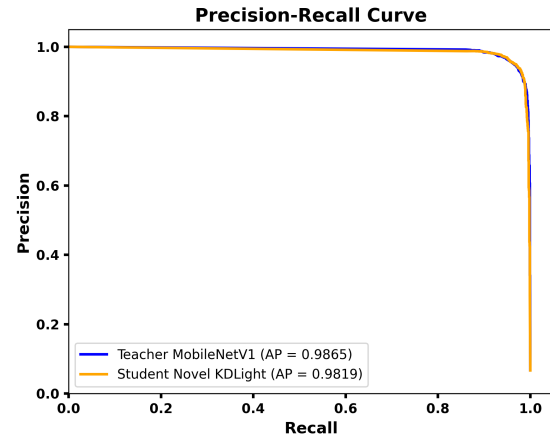


Fig. 5. Precision-recall curve visualization for multi medical disease

with ResNet152V2-Base, which achieves a higher recall rate of 93.64% and an accuracy of 94.48%. The performance of the remaining models falls within a recall range of 90.76% to 92.97% and an accuracy range of 93.09% to 94.35%. Despite the deeper and more complex architectures of these advanced teacher models, all of them demonstrate lower recall and accuracy compared to the proposed lightweight KDLight model. This highlights the ability of the KDLight model to outperform larger and more complex architectures while maintaining its efficiency and lightweight design.

TABLE VII
PERFORMANCE ANALYSIS OF NOVEL KDLIGHT STUDENT MODEL WITH DIFFERENT TEACHER MODELS THROUGH KNOWLEDGE DISTILLATION

Models	Precision	Recall	F1-Score	Accuracy
KDLight + DenseNet121	90.90	89.12	89.43	92.27 ↓
KDLight + DenseNet169	94.96	92.26	92.87	94.22 ↓
KDLight + DenseNet201	94.70	91.77	90.91	94.09 ↓
KDLight + InceptionV3	93.07	90.76	91.93	94.35 ↓
KDLight + ResNet101V2	93.10	92.97	92.26	93.14 ↓
KDLight + ResNet152V2	93.02	93.64	92.46	94.48 ↓
KDLight + Xception	92.18	92.16	92.62	93.09 ↓

V. CONCLUSION AND FUTURE WORK

In this paper, we introduced KDLight, a novel lightweight model for multidisease medical classification, leveraging Knowledge Distillation to enhance performance while maintaining low computational requirements. By transferring knowledge from the robust MobileNet teacher model, KDLight improves its ability to generalize across fifteen diverse medical conditions, achieving 95.55% accuracy, making it highly effective for edge devices with a compact storage requirement of 7.5 MB for generalized disease detection. Our approach not only demonstrates significant gains in accuracy and efficiency but also ensures that the model can operate effectively in resource-constrained healthcare environments. Additionally, KDLight's versatility extends beyond medical applications, offering promising potential for surface defect detection in various domains. The results highlight KDLight effectiveness in real-time, multidisease detection scenarios, emphasizing its practical relevance and broad applicability. In future work, we plan to explore the application of KDLight in engineering domains, such as crack versus non-crack detection, to further demonstrate its versatility and effectiveness in non-medical settings.

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